

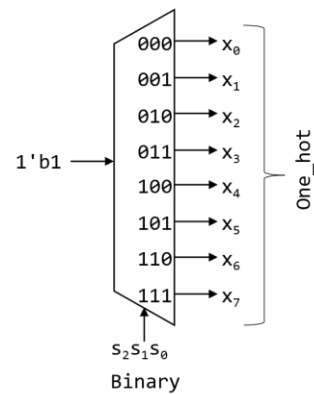
Homework 1

QUESTIONS

1.1 QUESTION 1

Model using HDL a look-up table **using a 1-to-16 de-multiplexer** that can act as a 4-bit one-hot (active-high outputs) decoder. The design should take as input a 4-bit binary value and output the one-hot equivalent. The definition of the module is given below. (Note: The image shows a 3-bit version)

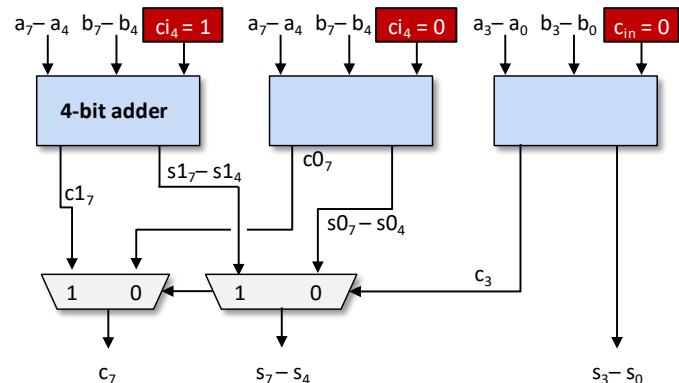
```
module decoder (  
    input  logic [3:0] binary,  
    output logic [15:0] one_hot  
);  
    // ...  
    // Add your description here  
    // ...  
endmodule
```



1.2 QUESTION 2

Model a design using HDL that implements an 8-bit unsigned carry select adder (CSA). The CSA should be built using 4-bit adders. The design of the adder is given to you. The definition of the module can be seen below.

```
module CSA_8 (  
    input  logic [7:0] A, B,  
    Input logic cin,  
    output logic [7:0] sum,  
    output logic carry  
);  
module adder_4 (  
    input logic [3:0] A, B,  
    input logic cin,  
    output logic [3:0] sum,  
    output logic carry  
);
```



1.3 QUESTION 3

- A) Model using HDL a unsigned N-bit multiplier. The design must be parametric with 2 inputs of N-bit, where $N > 3$. Your design should use half adders and/or full adders (given below).

```
module multiplier #(parameter N) (  
    Input logic [N-1:0] a,b,  
    output logic [2*N-1:0] product  
);
```

```
module full_adder (  
    Input logic a,b,c_in,  
    Output logic c_out, s  
);  
    logic s1,c1,c2;  
    half_adder ha1(a,b,s1,c1);  
    half_adder ha2(s1,c_in,c2,s);  
    assign c_out = c1|c2;  
Endmodule
```

```
module half_adder (  
    Input logic a,b,  
    Output logic c_out, s  
);  
    assign s = a^b;  
    assign c_out = a&b;  
endmodule
```